File Copy Studio

User Manual

Version 2026.05.09.0001

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File Copy Studio is a professional folder-to-folder file transfer utility for Windows. It offers configurable conflict resolution, CRC32 integrity verification, crash-safe resume, a built-in job scheduler, and full WCAG 2.2 AAA accessibility — all in a compact self-contained executable that requires no .NET runtime on the target machine.

1 Overview

File Copy Studio is designed for power users, IT professionals, and anyone who needs reliable, repeatable file transfers. Unlike a basic drag-and-drop copy, it gives you full visibility into every planned action before a single byte is written, and lets you tune every aspect of the transfer — parallelism, buffer size, bandwidth throttle, conflict strategy, and more.

Key capabilities at a glance:

- Conflict resolution with six modes: Overwrite, Overwrite-if-newer, Overwrite-if-older, Skip-if-same-date, Skip-all, and Rename
- CRC32 verification after every copy, or as a standalone verify-only pass
- Crash-safe resume: interrupted copies restart from the last 64 MB checkpoint
- Multi-threaded parallel transfers with automatic drive-affinity detection
- Built-in job scheduler with cron-syntax expressions
- Dry-run mode: plan and preview without touching the destination
- Bandwidth throttle in Mbps to protect shared network links
- Symlink policy: follow, skip, or copy as link
- 5 named presets for instant one-click job recall
- Portable mode for USB drives — no installer, no registry
- WCAG 2.2 AAA accessibility throughout

2 System Requirements

Requirement	Minimum	Recommended
OS	Windows 10 (1809+) x64	Windows 11 x64
.NET runtime	None — self-contained	None — self-contained
RAM	64 MB	256 MB+
Disk	~45 MB (app)	SSD recommended for large transfers
Display	1024×600	1920×1080 or higher

Note: No .NET runtime installation is required. The executable is fully self-contained.

3 Installation

3.1 Standard Installation

Run **FileCopyStudio_Setup.exe** and follow the on-screen prompts. The installer will place the application in *C:\Program Files\File Copy Studio*, create Start Menu and Desktop shortcuts, register the *.fclist* file type, and add shell context-menu entries to folders and drives.

3.2 Portable Installation

Extract **FileCopyStudio_Portable_<version>.zip** to any folder — including a USB drive. The archive contains:

- **FileCopyStudio.exe** — the application
- **portable.flag** — activates portable mode on startup
- **manual.pdf** — this document
- **Data** — created on first run; stores prefs.json and logs

In portable mode all settings and logs are written to the *Data*\ folder next to the executable — nothing is written to *%APPDATA%* or the registry. The title bar displays **[Portable]** to confirm the mode is active. Delete *portable.flag* to switch back to installed mode.

3.3 Silent / Scripted Installation

```
FileCopyStudio_Setup.exe /S
```

Installs silently to the default directory with no UI.

3.4 Uninstallation

Use *Add or Remove Programs* → **File Copy Studio**, or run *Uninstall.exe* from the installation folder. The uninstaller prompts whether to keep your saved presets and settings.

4 Interface Tour

The main window is divided into six horizontal zones. A resize grip in the bottom-right corner allows free window resizing.

Zone 1 — Identity / Scale

Shows the application name and a UI-scale selector (Compact / Standard / Medium / Large / Extra Large). Also displays the current theme and version. The scale selector resizes all controls proportionally for high-DPI displays or accessibility needs.

Zone 2 — Navigation Row

Contains all primary action buttons: Browse Source, Browse Destination, Swap Paths, Scan, Verify Only, Start, Pause/Resume, Cancel, Dry Run, Load Job, Save Job, Scheduler, and About. All buttons include keyboard shortcuts and ARIA labels.

Zone 3 — Path / Progress

Source and destination path text boxes with Browse buttons. Below them, two progress bars show current-file progress and overall transfer progress, with a live speed readout (e.g. 1.2 GB/s) and a sparkline chart of recent throughput.

Zone 4 — Options / Sliders

All copy settings: conflict mode drop-down, checkboxes for Subfolders / Timestamps / Attributes / Verify CRC / Move / Resume / Auto-parallel / Tray / Sound / Pause on battery, sliders for Buffer size, Parallel workers, and Throttle, and text fields for Include/Exclude masks and the Symlink policy drop-down. A summary line shows the scan result count at the bottom.

Zone 5 — File Grid

A sortable, filterable grid listing all planned file actions. Columns: Status icon, Relative path, Size, Source modified, Destination modified, Action, and Error. Right-click any row for context-menu options including Discard partial and Open in Explorer. The error log below the grid shows per-file errors with timestamps; errors can be copied to clipboard or cleared.

Zone 6 — Preset Bar

Five preset slots (1–5). A filled circle indicates the slot is populated. Click a slot to load its saved paths and options. Ctrl+Shift+1–5 saves the current settings to a slot. The Save Preset button opens a dialog to name the preset.

5 Quick Start

To perform a basic folder copy in four steps:

Step 1: Set paths

Click **Browse Source** (or type directly) to choose the source folder. Click **Browse Dest** to choose the destination folder. You can also drag-and-drop folders onto either path box.

Step 2: Scan

Click **Scan** (or press Ctrl+S). The engine enumerates both folders, compares files, and populates the file grid with the planned action for each file. Review the grid to confirm the planned actions before proceeding.

Step 3: Adjust options (optional)

In Zone 4, change the conflict mode, toggle Verify CRC, set parallelism, etc. After changing any option that affects planning, click Scan again to refresh the grid.

Step 4: Start

Click **Start** (or press F5). Progress is shown in Zone 3. Click **Pause** to suspend the transfer at any time; click it again to resume. Click **Cancel** to abort — already-copied files are kept.

> *Note: If the transfer is interrupted (power loss, crash), simply re-run. The next Scan will detect .fctemp partial files and mark them for Resume instead of full re-copy.*

6 Conflict Resolution

When a file already exists at the destination, File Copy Studio applies the configured conflict mode. The mode is set in the **On conflict** drop-down in Zone 4.

Mode	Description
Overwrite	Always overwrite the destination file.
Overwrite if newer	Overwrite only if the source is newer than the destination. Default.
Overwrite if older	Overwrite only if the source is older than the destination.
Skip if same date	Skip if source and destination have the same last-modified timestamp.
Skip all	Never overwrite; always skip existing files.
Rename	Copy and rename the source file (appends _1, _2, … until unique).

7 CRC32 Verification

File Copy Studio uses CRC32 (via System.IO.Hashing) to verify file integrity. Verification can be run in two ways:

- **Post-copy verify** — enable *Verify CRC* in Zone 4 before starting a copy. After each file is written, its CRC is computed and compared against the source CRC. Files that fail verification are marked Failed in the grid and logged.
- **Standalone verify** — click *Verify Only* (or press Ctrl+V) at any time. This reads every listed file from the destination and verifies it against the source without copying anything. Use this to audit a previous transfer.

7.1 CRC Sidecar Files

When **Write CRC sidecar** is enabled, a *.crc* file is written alongside each destination file after a successful copy (e.g. *photo.jpg.crc*). Subsequent verify-only passes use these sidecar files to confirm integrity without needing to read the original source.

8 Crash-Safe Resume

File Copy Studio writes a pair of resume marker files during every copy:

File	Purpose
.fctemp	The partial destination payload — raw bytes, no embedded header.
.fctemp.meta	JSON sidecar: source length, source mtime ticks, partial CRC32, byte offset.

Every 64 MB of written data, the meta sidecar is atomically updated (write-to-temp, rename). If the process is killed between updates, at most 64 MB of data is re-copied on resume. On the next Scan, the engine detects *.fctemp* files and marks those items *Resume* instead of *Copy*.

If the source file has changed since the partial was written (different size or mtime), the partial is automatically discarded and the copy restarts from the beginning.

> *Note: To discard a partial manually, right-click the file in the grid and choose Discard partial. Do not delete .fctemp files in Explorer while a copy is running.*

9 Parallel Transfers

File Copy Studio can copy multiple files simultaneously using worker threads. The optimal number of workers depends on your hardware:

Scenario	Recommended setting
Source and destination on the same spinning disk	1× (sequential — avoids head thrashing)
Source and destination on different disks	2–4× (auto-detected with Auto-parallel)
NVMe SSD source or destination	4–8× (storage is not the bottleneck)
Network destination (NAS, SMB share)	2–4× (depends on network bandwidth)

When **Auto-parallel** is enabled (default), the engine detects whether source and destination are on the same physical volume and sets parallelism to 1 if so, or to your configured maximum if they are on different volumes. Disable Auto-parallel to force a fixed thread count.

10 Job Scheduler

The built-in scheduler lets you automate recurring transfers without any external task scheduler. Open it with the **Scheduler** button in Zone 2 or press Ctrl+Shift+S.

10.1 Creating a Scheduled Job

In the Scheduler window:

- Click **New Job**.
- Enter a descriptive name.
- Set Source path and Destination path.
- Choose conflict mode, include/exclude masks, and any other options.
- Enter a schedule expression in the *Schedule* field (cron syntax — see below).
- Optionally set a per-job throttle in Mbps.
- Click **Save job**. The job appears in the list and is enabled immediately.

10.2 Cron Schedule Syntax

Schedule expressions use standard 5-field cron syntax:

```
MINUTE  HOUR  DAY-OF-MONTH  MONTH  DAY-OF-WEEK
  0–59   0–23      1–31      1–12     0–7 (0/7=Sun)
```

Expression	Meaning
0 2 * * *	Every day at 02:00
0 * * * *	Every hour (on the hour)
*/15 * * * *	Every 15 minutes
30 18 * * 5	Every Friday at 18:30
0 3 1 * *	First day of every month at 03:00
0 8,17 * * 1-5	Weekdays at 08:00 and 17:00

10.3 Managing Jobs

Jobs can be enabled or disabled individually using the toggle in the job list. Disabled jobs are retained but will not fire on schedule. To edit a job, select it in the list and modify the fields, then click Save job. To delete a job, select it and click Delete job.

> *Note: Scheduled jobs run inside the File Copy Studio process. The application must be running (or in the system tray) for scheduled jobs to fire. For fully unattended operation, configure Windows Task Scheduler to launch File Copy Studio at startup with the --minimized flag.*

11 Presets

Presets save a complete snapshot of your current settings — source path, destination path, conflict mode, include/exclude masks, all checkboxes, and buffer/parallel/throttle sliders. Up to five named presets are stored persistently in your preferences file.

Action	Method
Save preset	Click Save Preset (Zone 6), or Ctrl+Shift+1–5
Load preset	Click the slot button (Zone 6), or Ctrl+1–5
Rename preset	Hold Ctrl and click the slot, or use Save Preset dialog
Clear preset	Load the slot then click Save Preset and leave the name blank

12 Job Files (.fclist)

A job file saves the complete transfer configuration plus the planned file list as a JSON document with the *.fclist* extension. Job files let you define repeatable batch transfers outside the scheduler.

12.1 Save a job

After scanning, click **Save Job** in Zone 2. Choose a location and filename. The file includes paths, options, and the full planned action list — useful for auditing exactly what was (or will be) transferred.

12.2 Load a job

Click **Load Job** and select an *.fclist* file. The paths and options are restored. Click Scan to refresh the planned action list against the current state of the file system (the stored list in the file may be stale).

12.3 Shell integration

Double-clicking an *.fclist* file in Explorer opens it in File Copy Studio. If the application is already running, the file is forwarded to the existing instance via named pipe.

13 Command-Line Interface

Argument	Description
--source <path>	Set the source path on launch.
--dest <path>	Set the destination path on launch.
--scan	Trigger an automatic scan after launch.
<path> (positional)	Treated as --source.

Examples:

```
FileCopyStudio.exe D:\Photos
FileCopyStudio.exe --source D:\Photos --dest E:\Backup --scan
```

If File Copy Studio is already running, the new instance forwards its arguments to the running instance via named pipe and then exits immediately. The running window comes to the foreground and applies the new paths.

14 Keyboard Shortcuts

Shortcut	Action
Ctrl+S	Scan
F5	Start copy
Esc	Cancel
Ctrl+P	Pause / Resume
Ctrl+D	Dry run
Ctrl+V	Verify only
Ctrl+Shift+S	Open Scheduler
Ctrl+1 – Ctrl+5	Load preset slot 1–5
Ctrl+Shift+1 – 5	Save to preset slot 1–5
Alt+F4	Exit (or minimise to tray if enabled)

15 Accessibility

File Copy Studio is designed to meet WCAG 2.2 Level AAA throughout its interface.

Feature	Implementation
Contrast (SC 1.4.6)	All text ≥ 7:1 contrast ratio on both Dark and Light themes.
Non-text contrast (SC 1.4.11)	All icons, borders, and focus indicators ≥ 3:1.
Touch targets (SC 2.5.5)	All interactive controls ≥ 44px in both dimensions.
Focus indicators (SC 2.4.11)	2px Gold (dark theme) / Cyan (light theme) focus ring on all controls.
No colour-only info (SC 1.4.1)	Status icons use glyphs (■, ✓, ■) alongside colour.
Live regions (SC 4.1.3)	Status text and progress are exposed to screen readers via AutomationPeer.
High-contrast	Automatically detects Windows High Contrast and swaps to system palette.
UI scale	Five scale levels (Compact → Extra Large) for low-vision users.

16 Settings Reference

Setting	Default	Description
Subfolders	On	Include subdirectories in scan and copy.
Timestamps	On	Preserve source last-write timestamp on copied files.
Attributes	On	Preserve source file attributes (hidden, read-only, etc.).
Verify CRC	Off	Compute and compare CRC32 after each file copy.
Write CRC sidecar	Off	Write a .crc file alongside each copied file.
Move (delete src)	Off	Delete source file after a verified successful copy.
Resume partials	On	Detect and resume interrupted .fctemp files.
Auto-parallel	On	Automatically set thread count based on drive topology.
Parallel workers	1	Maximum number of concurrent copy threads (1–8).
Buffer size	1024 KB	Read/write buffer per worker (64 KB – 64 MB).
Throttle	0	Maximum bandwidth in Mbps (0 = unlimited).
Symlink policy	Follow	Follow / Skip / Copy-as-link.
Include mask	*.*	Semicolon-separated wildcard patterns to include.
Exclude mask	(empty)	Semicolon-separated wildcard patterns to exclude.
Minimize to tray	Off	Hide to system tray instead of taskbar on minimize.
Completion sound	On	Play a system sound when a transfer completes.

Setting	Default	Description
Open dest on done	Off	Open destination folder in Explorer after clean finish.
Pause on battery	Off	Automatically pause transfers when on battery power.
Theme	Dark	Dark / Light / Auto (follows Windows setting).
UI scale	Standard	Compact / Standard / Medium / Large / Extra Large.

17 Troubleshooting

Scan returns 0 files

Check that the source path exists and is accessible. If an include mask is set, ensure it matches the files you expect (default is *.*).

Files fail CRC verification

This indicates a read error or a write error during the copy. Retry the copy. If failures persist, check the source media for errors (run chkdsk or the drive's S.M.A.R.T. diagnostics).

Resume does not trigger

Resume only activates when Resume partials is enabled (Zone 4) and the .fctemp + .fctemp.meta pair is present and valid. If the source file has changed since the partial was written, the partial is discarded and a full copy is performed instead.

Scheduled job does not fire

The application must be running or minimised to the system tray for jobs to fire. Verify the schedule expression is correct. Check the log file in %APPDATA%\File Copy Studio\logs\ for any scheduler errors.

Shell context menu not appearing

Right-click a folder and look for 'Copy with File Copy Studio'. If missing, repair the installation by running the installer again. Context menus can take a few seconds to appear on first Explorer restart.

Application is very slow on a network share

Reduce parallelism to 1–2 workers. Increase buffer size to 4096 KB or more. Consider enabling throttle to avoid saturating the link.

High memory usage during large transfers

Each parallel worker allocates one buffer. With 8 workers and a 4096 KB buffer, peak allocation is ~32 MB from the pool. If memory is constrained, reduce parallel workers or buffer size.

18 Log Files

File Copy Studio writes structured logs to a daily rolling file:

Mode	Log location
Installed	%APPDATA%\File Copy Studio\logs\filecopy-studio-YYYY-MM-DD.log
Portable	.\Data\logs\filecopy-studio-YYYY-MM-DD.log (next to the exe)

Logs include all file-level operations, errors, scheduler events, and performance metrics. They are written in plain text and are safe to open in any text editor. Old log files are retained indefinitely — you may delete them manually at any time.

19 Build Notes

19.1 Prerequisites

Tool	Version	Purpose
.NET SDK	10.0	Compile and publish
NSIS	3.x	Build setup installer (optional)
Python	3.10+	Regenerate this PDF (optional)

19.2 Build commands

```
cd FileCopy
build.bat               # versioned installer + exe
build-portable.bat      # portable zip
```

build.bat auto-increments the build number (yyyy.MM.dd.NNNN), patches FileCopy.csproj and installer.nsi, publishes a self-contained single-file Release exe, copies README.md and manual.pdf to the dist folder, then runs makensis if NSIS is on PATH.

19.3 Project structure

File / folder	Purpose
FileCopy.csproj	Project file — .NET 10, WPF, self-contained publish
App.xaml / App.xaml.cs	Application entry, global styles, theme switching
MainWindow.xaml	Main window XAML layout
MainWindow.*.cs	Partial classes: Copy, Scan, Scheduler, Settings, UI
CopyEngine.cs	Core copy/verify engine with async parallel workers
ICopyEngine.cs	DI abstraction for unit testing
AppPrefs.cs	JSON-serialized user settings (Source Gen)
SchedulerWindow.xaml	Built-in job scheduler UI
Themes/Dark.xaml	Dark colour palette
Themes/Light.xaml	Light colour palette

File / folder	Purpose
PartialFile.cs	Resume marker read/write/finalize logic
DriveAffinity.cs	Drive topology detection for auto-parallel
SizeFormat.cs	Human-readable byte formatting
installer.nsi	NSIS installer script
build.bat	Build script (version increment + publish + installer)
build-portable.bat	Portable zip build script
manual.pdf	This document

20 Recent Version History

Version	Highlights
2026.05.09.0001	WCAG 2.2 AAA certification pass: BrushSelectedText on PrimaryButton/DangerButton, indeterminate CheckBox visual, BrushRed contrast correction. Performance: meta flush raised 16→64 MB, single FileInfo stat in Enumerate, CreateDirectory HashSet caching per group.
2026.05.08.x	SchedulerWindow WCAG fixes: Save job button contrast, 44px form controls, focus rings on LstJobs. Preset button Move left/right. BtnUiScale AutomationProperties.Name fix.
2026.05.07.x	Built-in job scheduler with cron expressions. Per-job throttle and conflict options. Tray minimize on scheduled run.
2026.05.04.x	Initial public release. Core copy engine, CRC32 verify, crash-safe resume, parallel workers, presets, job files, shell context menu integration.

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